

MEDGRID Development Phases

MEDGRID is a distributed healthcare resource exchange and emergency coordination platform. This document outlines the complete phased implementation roadmap for the MEDGRID system, starting from foundational architecture setup to enterprise-scale intelligent healthcare coordination.

Phase 0 — System Foundation & Architecture Validation

- Establish monorepo structure, TypeScript standards, CI/CD pipelines, and shared domain models.
- Define architecture decision records (ADRs) and environment configuration standards.
- Prepare database strategies and migration conventions.

Phase 1 — Core Monolith MVP

- Build initial centralized MVP with authentication, RBAC, inventory CRUD, and frontend dashboards.
- Validate healthcare workflows before introducing distributed systems complexity.
- Use modular architecture to preserve future microservice separation.

Phase 2 — API Gateway & Security Layer

- Introduce centralized JWT validation, RBAC enforcement, rate limiting, and request logging.
- Create unified entry point for all services.
- Implement suspension enforcement and security middleware.

Phase 3 — Facility Service Isolation

- Separate facility inventory management into isolated microservices.
- Introduce distributed database ownership per facility.
- Implement inventory summary synchronization.

Phase 4 — Geo-Discovery & Coordination

- Enable geographic facility discovery using Haversine distance calculations.
- Provide interactive map-based resource discovery using Leaflet and OpenStreetMap.
- Implement ranked discovery results.

Phase 5 — Resource Request Lifecycle

- Implement request creation, acceptance, rejection, and fulfillment workflows.
- Add reservation systems and audit logging.
- Introduce concurrency protection and transactional consistency.

Phase 6 — Expiry Intelligence & Redistribution

- Create expiry monitoring subsystem for near-expiry medical items.
- Implement redistribution offers and intelligent resource redistribution.
- Add optimistic locking and race-condition handling.

Phase 7 — EMCON Real-Time Operations

- Implement Emergency Medical Control (EMCON) live operational dashboards.
- Track ICU, ventilators, beds, ambulances, and emergency department statuses.
- Provide real-time hospital operational visibility.

Phase 8 — Reliability & Distributed Resilience

- Implement retry systems, exponential backoff, and health monitoring.
- Improve eventual consistency handling and distributed fault tolerance.
- Prepare the platform for real distributed deployments.

Phase 9 — Observability, Security & Production Hardening

- Introduce centralized monitoring, tracing, metrics, and security hardening.
- Implement audit logging, MFA groundwork, caching, and compliance tooling.
- Prepare the platform for production-grade deployment.

Phase 10 — Advanced Scaling & Intelligent Coordination

- Introduce event-driven architecture using Kafka, RabbitMQ, or NATS.
- Implement predictive analytics and intelligent healthcare allocation systems.
- Expand toward regional and enterprise-scale healthcare coordination.

Suggested Implementation Timeline

Timeline	Focus
Month 1	Phase 0 – 1
Month 2	Phase 2 – 3
Month 3	Phase 4 – 5

Month 4	Phase 6
Month 5	Phase 7
Month 6	Phase 8 – 9
Future	Phase 10

MEDGRID represents a highly ambitious and impactful distributed healthcare coordination platform. The phased implementation strategy ensures progressive validation, architectural stability, and manageable engineering complexity throughout development.